PRACTICE SHEET
ENGLISH FOR PROFESSIONAL PURPOSES II, GLA U
Semester IV **Module I: READING COMPRESHENION-III** **2026**

RC-I

Medical waste has been a growing concern because of recent incidents of public exposure to discarded blood vials, needles (sharps), empty prescription bottles, and syringes. Medical waste can typically include general refuse, human blood and blood products, cultures and stocks of infectious agents, laboratory animal carcasses, contaminated bedding material, and pathological wastes.

Wastes are generally collected by gravity chutes, carts, or pneumatic tubes, each of which has its own advantages and disadvantages. Chutes are limited to vertical transport, and there is some risk of exhausting contaminants into hallways if a door is left open during use. Another disadvantage of gravity chutes is that the waste container may get jammed while dropping, or it may be broken upon hitting the bottom. Carts are primarily for horizontal transport of bagged or containerized wastes. The main risk here is that bags may be broken or torn during transport, potentially exposing the worker to the wastes. Using automated carts can reduce the potential for exposure. Pneumatic tubes offer the best performance for waste transport in a large facility. Advantages include high-speed movement, movement in any direction, and minimal intermediate storage of untreated wastes. However, some objects cannot be conveyed pneumatically.

Off-site disposal of regulated medical wastes remains a viable option for smaller hospitals (those with less than 150 beds). Some preliminary on-site processing, such as compaction or hydropulping, may be necessary prior to sending the waste off site. Compaction reduces the total volume of solid wastes, often reducing transportation and disposal costs, but it does not change the hazardous characteristics of the waste. Compaction may not be economical if transportation and disposal costs are based on weight rather than volume.

Hydropulping involves grinding the waste in the presence of an oxidizing fluid, such as hypochlorite solution. The liquid is separated from the pulp and discharged directly into the sewer unless local limits require additional pretreatment prior to discharge. The pulp can often be disposed of at a landfill. One advantage is that waste can be rendered innocuous and reduced in size within the same system. Disadvantages are the added operating burden, difficulty of controlling fugitive emissions, and the difficulty of conducting microbiological tests to determine whether all organic matters and infectious organisms have been destroyed from the waste.

On-site disposal is a feasible alternative for hospitals generating two tons or more per day of total solid waste. Common treatment techniques include steam sterilization and incineration. Although other options are available, incineration is currently the preferred method for on-site treatment of hospital waste.

Steam sterilization is limited in the types of medical waste it can treat, but is appropriate for laboratory cultures and/or substances contaminated with infectious organisms. The waste is subjected to steam in a sealed, pressurized chamber. The liquid that may form is drained off to the sewer or sent for processing. The unit is then reopened after a vapor release to the atmosphere, and the solid waste is removed for further processing or disposal. One advantage of steam sterilization is that it has been used for many years in hospitals to sterilize instruments and containers and to treat small quantities of waste. However, since sterilization does not change the appearance of the waste, there could be a problem in gaining acceptance of the waste for landfilling.

A properly designed, maintained, and operated incinerator achieves a relatively high level of organism destruction. Incineration reduces the weight and volume of the waste as much as 95% and is especially appropriate for pathological wastes and sharps. The most common incineration system for medical waste is the controlled-air type. The principal advantage of this type of incinerator is low particulate emissions. Rotary-kiln and grate-type units have been used, but use of grate-type units has been discontinued because of high air emissions. The rotary kiln also puts out high emissions, and the costs have been prohibitive for smaller units.

1. Which of the following organizational schemes is most prevalent in the passage?
a. chronological order
b. comparison-contrast
c. order by topic
d. hierarchical order

2. One disadvantage of the compaction method of waste disposal is that it
a. cannot reduce transportation costs.
b. reduces the volume of solid waste material.
c. does not allow hospitals to confirm that organic matter has been eliminated.
d. does not reduce the weight of solid waste material.

3. For hospitals that dispose of waste on their own premises, the optimum treatment method is
a. incineration.
b. compaction.
c. sterilization.
d. hydropulping.

4. According to the passage, which of the following could be safely disposed of in a landfill but might not be accepted by landfill facilities?
a. hydropulped material
b. sterilized waste
c. incinerated waste
d. laboratory cultures

5. The two processes mentioned in the passage that involve the formation of liquid are
a. compaction and hydropulping.
b. incineration and compaction.
c. hydropulping and sterilization.
d. sterilization and incineration.

6. According to the passage, two effective methods for treating waste caused by infectious matter are
a. steam sterilization and incineration.
b. hydropulping and steam sterilization.
c. incineration and compaction.
d. hydropulping and incineration.

7. Hospitals can minimize employee contact with dangerous waste by switching from
a. a manual cart to a gravity chute.
b. an automated cart to a hydropulping machine.
c. a gravity chute to a manual cart.
d. a manual cart to an automated cart.

8. The process that transforms waste from hazardous to harmless and diminishes waste volume is
a. sterilization.
b. hydropulping.
c. oxidizing.
d. compacting.

9. The underlined word exhausting, as it is used in the second paragraph of the passage, most nearly means
a. debilitating.
b. disregarding.
c. detonating.
d. discharging.

10. Budgetary constraints have precluded some small hospitals from purchasing
a. pneumatic tubes.
b. rotary kilns.
c. sterilization equipment.
d. controlled-air kilns.